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What Factors are Relevant to Understanding Dropout? Analysis at a Co-Financed University in Ecuador and Policy Implications, Using Survival Cox Models

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ABSTRACT
University dropout is a serious problem in higher education that is increasingly gaining importance, as it is essential to understand its causes and search for public and institutional policies that can help reduce it. This research uses conventional and extended Cox survival models to analyze the factors behind dropout rates at a co-financed higher education institution (HEI) in Ecuador. The latter allows us to model specific components, such as time-dependent and independent variables, and unobserved heterogeneity. The results show that certain student-related, family-related, or background characteristics are relevant for the ecuadorian context. However, the most influential factors affecting the dropout risk are institutional and academic. These findings provide valuable insights into the actions and policies that HEI can implement to mitigate dropout rates through internal measures that public policies can complement.

KEYWORDS
Higher education dropout; Latin America; factors dropout; survival models; extended cox model; time-dependent covariate

Introduction

Dropout is one of the phenomena that most affects higher education worldwide, generating consequences in the economic, social, and personal aspects of those who abandon their studies. According to the International Institute for Higher Education in Latin America and the Caribbean (UNESCO-IESALC, 2020), dropout rates could reflect the ineffectiveness of higher education systems: in other words, guaranteeing access to education is not enough; student graduation also needs to be prioritized to ensure qualified human capital that contributes to society.

The OCDE (2022) specifies that in its member countries, on average, 12% of those who enroll in undergraduate studies drop out before starting their second year. Additionally, 21% do not graduate within the theoretical duration of the program, and that figure rises to 23% when considering an additional 3 years for graduation. These figures can be more critical in some countries; for example, the proportion of students who do not complete their undergraduate studies within the theoretical time plus 3 years is over 40% in countries such as Brazil, Colombia, Italy, Slovenia, Austria, and Switzerland. In the U.S., the Admissionsly (2020) reveals that university dropout rates average around 40%, and the majority drop out in the first year; thus, approximately one out of every three students who enroll in higher education in the U.S. never manages to obtain a degree. In their New York Times article, Leonhardt and Chinoy (2019) points out that the problem of lack of completion in studies is a central concern for administrators and policymakers in the U.S.; therefore, organizations like

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Alliance (a network of charter schools in Los Angeles) have started to delve deeper into the problem, finding that, on average, universities have lower graduation rates when they enroll more low-income students, more African American and Latino students, more men, more mature students, and more students with low SAT or ACT scores. On the other hand, in regions like Latin America, the World Bank (2017) indicates that approximately twenty million young people enrolled in Higher Education Institutions in the last decade, but only half obtained a degree.

In Ecuador, education is recognized as a fundamental right for all citizens, positioning it as a high-priority sphere in public policy and state investment. This approach ensures equity, social inclusion, and societal well-being (Constitución de la República del Ecuador, 2008, Article 26). The higher education system in Ecuador is structured into three main categories: universities, polytechnic schools, and technical and technological institutes. When delving into the realm of universities, the Ecuadorian system encompasses three different types of institutions: public, co-financed, and private, serving 59.84%, 26.65%, and 13.51% of students, respectively. Regarding the university dropout rate, official figures from 2019 indicate an average rate of 18.39% for students who drop out in the first year, showing an improvement compared to the 26.30% recorded in 2015.Interestingly, universities that receive co-financing from the government exhibit the highest dropout rate, standing at 29.7% (SENESCYT, 2022).

The phenomenon of permanent dropout, that is, when students permanently leave the higher education system (Vergara et al., 2017), has negative consequences for students as well as institutions and society in general. According to UNESCO-IESALC (2007), dropping out has various effects, such as feedback into the cycle of poverty, increased unemployment, dependence on government assistance, decreased efficiency and quality indexes of higher education institutions (HEIs), and economic and financial implications for HEIs. Therefore, it is unsurprising that this phenomenon is taken into account in the allocation of government resources to universities and in the creation of different accreditation models and international rankings.

Dropout is a multicausal problem that involves structural (access versus graduation), institutional, and individual factors, among others. Despite the diversity of causes, there is consensus that the university dropout usually occurs during the first years (Arias Ortiz & Dehon, 2013; Juajibioy, 2016; Osorio et al., 2012; Tinto, 1975). This relationship between dropout risk and elapsed time of studies can be modeled using survival analysis, which is a statistical method that models the time until an event occurs.

Survival analysis also makes it possible to model the presence of censored observations. Among the different types of censoring, the most common is "right censoring," which occurs when an individual does not show the event of interest during the analysis time; therefore, his/her last observed follow-up time is less than the time that can elapse until the event occurs. For Willett and Singer (1991), censored observations create difficulties in the estimation and inference of parameters when using others methods.

Among the survival analyses, one of the most frequently used methods is the Cox model because it assumes that each individual's hazard rate is proportional to that of other individuals at a constant rate over time; as such, it is also known as the "Cox proportional hazards model" (Ratnaningsih et al., 2021). In practice, this assumption is not always correct, particularly when longitudinal and covariate data that vary over time are used, which would produce nonproportional hazard rates among individuals. When the assumption of proportional hazards is violated, extended Cox models should be used, which allow time-dependent variables to be included.

Survival analysis has been applied to study higher education dropout; however, extended Cox models have been less frequently used in this type of analysis. Furthermore, with Ecuador specifically, no studies have been found that utilize survival models to study dropout. The few studies in Ecuador that have been conducted, have focused on descriptive analysis (see: Rubio Gomez et al., 2012; Sinchi & Gómez, 2018; Viteri & Uquillas, 2011; Zambrano Verdesoto et al., 2018); however, in recent years, some authors (Alban & Sanchez, 2018; Heredia-Jimenez et al., 2020; Sandoval-Palis et al., 2020) have

delved deeper into the subject; their analysis have centered on the creation of predictive models, which disregard the moment when the event occurs.

Thus, using survival models (the conventional Cox model and extended Cox model), the current study aims to explore which factors – individual, family, institutional, and academic – are associated with university dropout. A case study is thus utilized to analyze student dropout in a co-financed University such as Pontificia Universidad Católica del Ecuador (PUCE-Q). This study contributes to the literature by applying a model that better represents the real conditions in which dropout occurs using survival models that include time-dependent and time-independent variables, as well as elements of heterogeneity. It also applies this method to the rarely studied context of Ecuador and a type of university where public and private resources are mixed.

Literature review

Relevant studies have shown that analyzing the dropout phenomenon is complex because of the multiple perspectives (individual, institutional, and national) involved (Tinto, 1975). Also, to carry out an adequate study, it is necessary to define and distinguish the type of dropout that will be analyzed. Dropout in general terms is defined by UNESCO (2012) as pupils who either no longer attend school, have moved to another school system, or have died. However, there is not a universally agreed-upon classification, as different authors may approach the topic from various perspectives. While the concepts of different types of university dropouts may not be explicitly mentioned in the theories, authors may consider some general categories, such as academic failure, transfers, permanent dropout, or temporary withdrawal (Hackman & Dysinger, 1970; Tinto, 1975). A transfer is considered when students pursuing a bachelor's degree change post-secondary institutes and/or programs before graduating (Fauria & Fuller, 2015). Failure is all those students who leave the university because they did not meet all passing requirements. These students are either dismissed for academic difficulty or withdrawn before official action. On the other side, withdrawal-returnees or temporary dropouts are those students in good standing who withdrew during a period of time but then returned (during the observed time). Finally, permanent dropouts or withdrawals are the students who left the university even if they met all passing requirements and had not returned in the observed time (Rossmann & Kirk, 1970).

The student attrition phenomenon has been a major concern for educational institutions and educators. Before 1970, their focus was on the characteristics of individual students rather than on their interactions with college environments. During the late 1960s and the 1970s, theoretical models were developed, and systematic studies and attempts to conceptualize retention frameworks that included the notion of the student – college relationship became more common (Aljohani, 2016). One of the first contributions was Spady's empirical model, which assess the independent contribution of the social process (family and previous educational background, academic potential, normative congruence, institutional commitment, etc.) over dropout (Spady, 1970). Other important theoretical models was Tinto's model of dropout, which explains that the withdrawal process from higher education can be viewed as a longitudinal process of interactions between academic and social systems that are continually modified by variance in the individuals' performance (Kerby, 2015).

Given the different theoretical approaches, we are interested in investigating the effect that the characteristics of the student and his family, institutional characteristics, and academic performance have on dropout.

From an empirical perspective, various methodologies have been used to analyze university dropout, but survival models have been preferred because they can be adjusted to model censored data, which occurs in student dropout. Willett and Singer (1991) conducted one of the first analyses on dropout using survival models; in their study, they point out that the methodology allows for a longitudinal analysis that considers censored data.

Several studies that use Cox models have found it occurs in the first two years of university (Alvarez, 2016; Paura & Arhipova, 2014; Sosu & Pheunpha, 2019), regarding the importance of the

time when dropout occurs. Student characteristics like sex, age, and high school grades are crucial factors when analyzing dropout; however, the results vary depending on the framework. For example, for Paura and Arhipova (2014), Alvarez (2016), and Da Costa et al. (2018), being male is an element of risk, while for Juajibioy (2016), the most significant risk is in female students. Eshghi et al. (2011) found age affects the decision to drop out, they concluded older students had a higher dropout risk. Grades from high school or the educational system have been found to be an important predictor of dropout, as students with a lower grade point average (GPA) have shown a greater probability of dropout (see: Eshghi et al., 2011; Paura & Arhipova, 2014; Sosu & Pheunpha, 2019).

When time was incorporated in Cox models, Lassibille and Gómez (2008), Ameri et al. (2016), and Ratnaningsih et al. (2019) found that dropout risk is higher for students with a lower level of formal education or who were older when starting or postponed university; additionally, Lassibille and Gómez (2008) found an association between dropout and students with a low socioeconomic level and insufficient financial support. Furthermore, Ameri et al. (2016) found that number of family members and ethnicity have a significant impact on dropout. On the other hand, Gury (2011) findings that aspects related to parents' education and working conditions also influence dropout. He also found that students from schools that receive public funding had a lower dropout risk and that those living with their parents during the first two years of school had a higher probability of dropping out.

Finally, some researchers like Wallner and Nissen (2018) and Ratnaningsih et al. (2021) have used survival models that incorporate aspects such as time and data heterogeneity (frailty), but although Wallner and Nissen (2018) found that sex did not affect dropout decision, Ratnaningsih et al. (2021), in line with previous results, determined women had a higher dropout risk, as did older, and low-GPA students.

It is also important to note a brief review of literature on the policies focused on reducing the impact of dropouts in HEIs, Tenório de Freitas and Bezerra (2022) identify four types: economic, assistance-based, academically focused, and criteria-based on students' needs/vulnerability, which can be implemented at both the government and HEI levels. The most common ones are scholarships, grants, and loans, which have a significant impact on the decisions to persist and complete studies (Eather et al., 2022; Tenório de Freitas & Bezerra, 2022). For example, Ison (2021) indicates that an outstanding tuition balance drastically decreases the likelihood of graduation. These types of policies are the most commonly implemented by governments. For example, in the Latin American context, the Ecuadorian government has a program to assign scholarships every year to students without economic resources or who belong to vulnerable groups through co-financed HEIs as PUCE, in the same way Colombian and Brazilian governments have several initiatives to provide financing to students (Arias et al., 2023; Tenório de Freitas & Bezerra, 2022)

In terms of policies focused on academics and the specific needs of students or their vulnerability, there are multiple actions implemented by HEIs (Arias et al., 2023) which range from leveling, accompaniment throughout student life, curricular reforms, to the offer of specific assistance services. Their results have depended largely on the context on which each of them is developed, finding examples in countries as US, where Bettinger and Baker (2014) research had demonstrated that programs of coaching mentorship have a positive effect in retention rates, along the same lines Canada has programs like peer support that have positive effects on student learning (Neitherman et al., 223). Latin American countries such as Bolivia, Brazil, Ecuador and México have mostly specific services to students that have been considered as good practices for their results (UNESCO, 2009)

Methodology

Context

In Ecuador, only 36.63% of individuals aged 18 to 24 have access to higher education, with 29.9% enrolled explicitly in a college. The total enrollment records establish that approximately 87% choose in-person mode for their studies, while the remaining 13% maintain distance, semi-presential or online modalities. As reported by SENESCYT (2020), students show a strong preference for fields such as social sciences, journalism, information, and law,

constituting a combined 34.8% of total registrations. Health-related programs attract 19% of enrollments, while engineering programs account for 15% of the student population. In terms of gross enrollment rates within the 18 to 24-year-old population, women have a higher percentage than men (39.83% and 33.52%, respectively). However, significant disparities exist at the geographical level, particularly in regions with higher poverty levels, such as certain provinces in the central highlands, coast, and eastern regions, where enrollment rates fall below 25%.

In this study, we have gathered information from the Pontifical Catholic University of Ecuador, Quito (PUCE-Q), a co-financed institution. These types of universities, known as co-financed, receive funds from the government to provide scholarships and financial assistance to students in need. As was previously indicated, only 13.3% of universities in Ecuador fall within this category (SENESCYT, 2020), yet 26.7% of the current student body is housed at these schools. In 2019, 18.38% of students discontinued their studies in their first year. Notably, universities co-financed by the government exhibit the highest dropout rate, representing 5.4% of the whole population (SENESCYT, 2022).

Data collection

The study population corresponds to the total of PUCE-Q students who entered the institution for the first time between September 2014 (cohort 2014–02) and February 2016 (cohort 2016–01), all of them were observed until the end of the semester. 2019–01. It is important to mention that the most of the degree programs included in the study last for eight semesters. Individuals in the sample were observed for a maximum of 10 semesters (for those who began in 2014–02) to minimum of six semesters (for those who began in 2016–01). Although not all students are observed for the same time interval, it is precise because of this limitation that survival models have advantages over other alternatives.

The data used were taken from records from university administration and government sources specifically, the Ecuadorian Ministry of Education to complete missing data about the type of High School and the Civil Registry Office to incorporate educations parents to university administrative records and added sociodemographic variables for students and their parents.

Regarding study population, of the 15 existing schools, the School of Medicine was excluded because it has a different system for student admission, monitoring, and evaluation from the rest of the university. Students from the Department of Education were also excluded from the analysis because of the prevalence of distance learning for this department. Additionally, students in the social work degree program and the Department of Philosophical and Theological Sciences were excluded because of low enrollment. Thus, of the 3,463 students who started at PUCE-Q from 2014–02 to 2016–01, the final sample included 2,472 students.

Data analysis

For this analysis, dropout was defined as the departure or exit from the institution by a student who, having enrolled in PUCE-Q, withdraws, either voluntarily or not. Therefore, if a student temporarily stopped enrolling (for one or more periods during the study period), but re-enrolled at or before 2019–01 (the last period for which information is available), he or she is not considered to have dropped out; similarly, if a student decided to change degree programs, he or she is not considered to have dropped out. Based on the described considerations, Table 1 shows how dropout occurred over time for the analyzed sample.

Of the 2,472 students, 516 dropped out, which constitutes approximately 21% of those who enrolled between 2014–02 and 2016–01. As in many related studies, the data show that most dropouts occurred in the first years of study. Thus, 43% of students completed only one semester and then dropped out, and 63% completed a maximum of two semesters and then left permanently; after four semesters, 88% of dropout students had already abandoned the institution.

Table 1. Number of dropouts per semester.

Number of enrolled semesters	Number of dropouts	Dropout (%)
1	222	43%
2	105	20%
3	87	17%
4	43	8%
5	32	6%
6	14	3%
7	13	3%
8	0	0%
9	0	0%
10	0	0%
Total	516	100%

Source: Prepared by the authors based on university administrative records.

Table 2. Descriptive statistics.

Observed Covariates	Category	Non-Dropouts	Dropouts	Total
Sex	Men	764	257	1021
	Women	1192	259	1451
Type of High School	Foreign/Unknown	23	20	43
	Private	1317	334	1651
	Public	339	94	433
	Semi-Public	277	68	345
Pichincha	Another Province	244	80	324
	Pichincha	1712	436	2148
Age at admission	≤18 years old	929	195	1124
	>18 years old	1027	321	1348
Mother's Education Level	Unknown	86	55	141
	Primary/Secondary	957	277	1234
	University	913	184	1097
Father's Education Level	Unknown	125	72	197
	Primary/Secondary	789	229	1018
	University	1042	215	1257
Cohort	2014–02	759	202	961
	2015–01	307	103	410
	2015–02	649	138	787
	2016–01	241	73	314
Discount Dummy	Does Not Receive Discount	124	55	179
	Receives Discount	1832	461	2293
Academic Department	Business	253	65	318
	Architecture	197	90	287
	Human Sciences	230	40	270
	Communication	181	33	214
	Economics	133	32	165
	Nursing	250	61	311
	Natural and Exact Sciences	60	12	72
	Engineering	197	104	301
	Law	169	30	199
	Psychology	286	49	335
Entrance Exam Score	Average	42.910	4.060	42.710
	Standard Deviation	7.897	7.218	7.885
Grade Point Average	Average	38.75	31.69	37.22
	Standard Deviation	3.82	6.35	5.48
Ratio of Failed Credits	Average	0.100	0.36620	0.1819

Source: Prepared by the authors using university administrative records and those from the Ministry of Education and Civil Registry Office.

The description of the main variables used for the analysis is presented in Table 2, divided between dropout and non-dropout students.

The most of PUCE-Q students were women, which also was the group with the fewest dropouts. As for type of high school, 47% of students from a foreign or unknown high school dropped out. Students living in

provinces other than Pichincha[1] were more likely to drop out (25%) than those living in Pichincha (20%). Further, 55% of students were 18 years old or younger when they enrolled, of which 17% dropped out; for students over 18 years old, 24% dropped out. Regarding parents' education level, students whose mothers had a primary/secondary-level education were the most likely to drop out, while for father's education level, the university-level category corresponded to the greatest number of dropouts.

In terms of institutional-related variables, 74% of students received discounted tuition from PUCE during their studies, either in the form of financial aid or a scholarship. The group that did not receive a discount was small but had a higher probability of dropout (31%). Both the architecture and engineering departments had the highest dropout rates (32% and 35%, respectively), while the psychology and human sciences departments had the lowest dropout rates (14.6% and 14.8%, respectively).

The average score on the admission test, which measures the student's abilities and aptitude, was 43. The average admission test score was 40 for dropout students and 43 for non-dropout students.

In terms of academic performance, which is related to GPA, the sample average was 37 points, 32 for dropout students, and 39 for non-dropout students. Finally, the ratio of failed courses was 0.1 for non-dropout students and 0.37 for dropout students.

Method of analysis

The Cox proportional hazards model, developed by Cox in 1972, was used. This model considers the hazard function h(t) based on time t and a set of covariates X. Following Boj Del Val (2017), the model is as follows:

$$h(t,X) = h\left(t, X_1, \ldots, X_p\right) = h_0(t) exp\left[\sum\nolimits_{j=1}^{p} \beta_j X_j\right] \tag{1}$$

Equation (1) has two components: that which depends on time, $h_0(t)$, also referred to as the "baseline hazard function," which does not consider covariates or regressors, and $exp\left[\sum_{j=1}^{p} \beta_j X_j\right]$, which solely depends on the regressor vectors $\left(X_1, \ldots, X_p\right)$, which are assumed to be independent of time. The result is a "semi-parametric" model. Once the parametric segment, corresponding to $exp\left[\sum_{j=1}^{p} \beta_j X_j\right]$, and then the nonparametric segment $h_0(t)$, are calculated, the complete model is written as:

$$\hat{h}(t,X) = \hat{h}_0(t) exp\left[\sum\nolimits_{j=1}^{p} \hat{\beta}_j X_j\right]. \tag{2}$$

Equation (2) allows the relationship between the dropout hazards of two students exposed to different risk factors to be determined. To this end, the model is based on a fundamental hypothesis where the existence of proportional risks is assumed. Determining the hazard ratio (HR) between two students is done using different vector of covariates, that is, $X = \left(X_1, \ldots, X_p\right)$ and $X^* = \left(X_1^*, \ldots, X_p^*\right)$, where X* represents the greatest risk and X the least risk; the comparison of both individuals can be expressed by:

$$HR = \frac{h(t,X^*)}{h(t,X)}. \tag{3}$$

Substituting (1) for (3) results in:

$$HR = \frac{h_0(t) exp\left[\sum_{j=1}^{p} \hat{\beta}_j X_j^*\right]}{h_0(t) exp\left[\sum_{j=1}^{p} \hat{\beta}_j X_j\right]} = exp\left[\sum\nolimits_{j=1}^{p} \hat{\beta}_j (X_j^* - X_j)\right], \tag{4}$$

[1]Pichincha province is used as the reference since it is where PUCE-Q is located. Most students from other provinces live alone or with relatives. PUCE-Q does not have a housing campus.

where HR does not depend on the baseline hazard function $h_0(t)$ since time-independent covariates are assumed. In this set of variables, variables such as sex, high school of origin, and parents' education can be represented.

However, given the possibility that the proportional hazards assumption may not be met for all covariates (for example, those of academic performance that may vary over time), the original model can be expanded to the Cox extended model. One extension involves including time-dependent covariates. Thus, the previous model can be generalized, considering the coexistence of two groups of covariates, some time-independent and others time-dependent. Therefore, the vector of both groups of covariates can be represented as $X(t) = \left(X_1, \ldots, X_{p1}, X_1(t), \ldots, X_{p2}(t)\right)$, so the Cox model would be:

$$\hat{h}(t, X(t)) = \hat{h}_0(t) exp\left[\sum_{j=1}^{p1} \hat{\beta}_j X_j + \sum_{j=1}^{p2} \hat{\delta}_j X_j(t)\right]. \tag{5}$$

This model considers variables that can change over time (Mills, 2011), such as semester grades and number of credits earned and failed, among others. One of the main modifications to the original Cox model is that the covariates depend on time; therefore, $X(t)$ represents the value of a covariate vector for individual j in time interval t.

Lastly, to model the group effect, the shared frailty model was used, which models the random effects in survival models (Cleves et al., 2010). This model was used because there may be a potential correlation between students induced by a latent effect at the department or degree-program level. This latent effect is also known as unobserved heterogeneity.

Thus, the model incorporates unobserved heterogeneity in Equation (6) through a multiplicative effect on the hazard function, where $i = 1, 2, \ldots n$ groups with $j = 1, 2, \ldots n_i$ members of group i.

Therefore, considering both shared frailty and the presence of time-dependent and independent covariates, the hazard function for the jth subject of the ith group is:

$$\hat{h}_{ij}(t, X(t)) = \hat{h}_0(t) \alpha_i exp\left[\sum_{j=1}^{p1} \hat{\beta}_j X_j + \sum_{j=1}^{p2} \hat{\delta}_j X_j(t)\right]. \tag{6}$$

where α_i represents frailty at the group level. The frailties are unobservable positive values that have a mean of 1 and a variance of θ, which are estimated from the data. For $v_i = log\alpha_i$, the hazard function is:

$$HR = exp\left[\sum_{j=1}^{p1} \hat{\beta}_j X_j + \sum_{j=1}^{p2} \hat{\delta}_j X_j(t) + \hat{v}_i\right]. \tag{7}$$

Findings

From Equations (4) and (7) and the data described in the previous section, calculations from two models were conducted; the results are presented in Table 3. Column 2 indicates the reference category used for the calculations. The other columns show the estimated HR with its level of significance and standard deviation for each model. It should be noted that HR values of less than 1 decrease dropout probability, while values greater than 1 increase dropout probability.

Model 1 corresponds to the Cox proportional hazards model, in which data pertaining to student and parent variables are used (i.e., pre-university admission characteristics). Variable-level and overall proportionality is also assumed in this model (Appendix, Table A1). On the other hand, Model 2, in addition to the variables in Model 1, incorporates institutional and academic variables (i.e., those that vary from semester to semester for each student and those that do not); therefore, Model 2 corresponds to the extended Cox model and combines time-dependent and independent covariates. In addition, the model incorporates random effects, which capture the unobserved correlation shared by

Table 3. Estimation of coefficients for the Cox proportional hazards model and extended Cox model with shared frailty.

		Model 1		Model 2	
		Cox proportional hazards		Extended Cox with time-dependent variables and shared frailty	
	Reference category	Hazard ratio	SD	Hazard ratio	SD
Student and parentcharacteristics					
Sex (Woman)	Man	0.71***	0.063	1.004	0.098
Age (≤ 18 years old)	>18 years old	0.791**	0.074	0.932	0.091
Origin (Pichincha)	Another province	0.938	0.126	0.925	0.125
High school (unknown/foreign)	Public high school	2.231***	0.553	2.078***	0.547
High school (private)		1.071	0.130	1.101	0.136
High school (semi-public)		0.973	0.158	0.97	0.158
Father's education (unknown)	Primary/Secondary	1.453*	0.284	1.29	0.262
Father's education (university)		0.825*	0.083	0.892	0.092
Mother's education (unknown)	Primary/Secondary	1.246	0.275	1.057	0.246
Mother's education (university)		0.77***	0.077	0.77***	0.786
Institutional and academic					
Cohort (2015–01)	Cohort 2014–02			0.814	0.105
Cohort (2015–02)				0.727***	0.082
Cohort (2016–01)				0.702**	0.105
Discount (Yes)	No discount			0.608***	0.091
Entrance exam score				0.977***	0.007
Grade point average (t)				0.98***	0.004
Ratio (failed credits/registered credits) (t)				2.1***	0.142
theta				0.046***	0.03
Number of students		2472		2472	
Number of observations		2472		17539	
LR Chi-square/Wald Chi-square		82.28***		775.64***	

***$p < .01$; **$p < 0.05$; *$p < 0.1$.
Source: Prepared by the authors using university administrative records and those from the Ministry of Education and Civil Registry Office.

individuals belonging to the same group – in this case, the same academic department. Both models are significant at the 1% level according to their goodness-of-fit values, measured using the LR chi-square and Wald chi-square statistics, respectively.

The Model 1 show that women have an HR of 0.71, meaning they have 29% less risk of dropping out compared to men. Regarding enrollment age, students who started university immediately after finishing high school (17 or 18 years old), had an HR of 0.79, indicating they had 21% less dropout than those who started when they were over 18. In terms of place of origin, although students from Pichincha province had an HR of 0.938 (i.e., 6.2% less risk of dropping out), this effect was not statistically significant; therefore, leaving another province to study at PUCE-Q was not related to dropout. With respect to students' high school, no statistically significant differences were observed between public schools (reference category) and private and semi-public schools. However, for unknown/foreign schools, there was a significant difference in dropout risk, which increased by 123% compared to public schools. Regarding parents' education had an important influence on dropout rate: the results show that a university-level education for the father and mother decreased dropout risk by 27.3% and 29.8%, respectively, compared to a primary/secondary-level education.

On the other hand, focusing the analysis on the results of Model 2, the only Model 1 variables that remained significant were unknown/foreign high school and mother's college education; the other variables lost their statistical significance. This shows that adding institutional and academic factors allowed observable student factors to be controlled for, which led to improved estimators. Further, the significance of the unknown/foreign high school variable may be related to the possibility that students obtained a degree abroad and, consequently, that relevant differences in academic levels and/or expectations exist. Regarding mother's university-level education, its significance could mean this variable exerts an economic and motivational influence.

The institutional and academic variables included in the model were highly significant. As such, the university admission cohort was a determinant for dropout: students who began university in the 2015–02 and 2016–01 cohorts had lower dropout risk (17.8% and 30%, respectively) than those from the 2014–02 cohort. Additionally, students who had received some type of tuition discount (related to academic, cultural, sports, or socioeconomic factors) were 39.2% less likely to drop out than those who did not receive any benefit. With respect to the university entrance exam score showed that one additional point decreased dropout risk by 2.3%. Regarding the academic performance variables, one point higher than mean GPA decreased the risk of dropping out by 2%, while one point higher for the ratio of failed credits versus registered credits, increased the risk of dropping out by 110%

Finally, about the shared frailty that allows the random effects to be modeled, the theta parameter of 0.046 is significant at 1%, so the presence of unobserved heterogeneity between the groups is supported, and the presence of frailty captured through faculty or school should be used in the model.

Discussion and policy implications

Previous results show that personal, family, and background variables influence dropout rates less than institutional and academic variables. This is a crucial initial finding since universities have limited control over students' pre-entry variables (background variables). Still, they could significantly impact dropout rates through institutional and academic policies.

Despite this, it can be helpful to keep an eye on some personal, family, and background variables, as they seem to influence dropout rates, not only in this study but also in other contexts. For instance, findings from (Paura & Arhipova, 2014; Alvarez, 2016; Da Costa et al., 2018; among others) indicate that women have a lower risk of dropping out compared to men. Severiens and Dam (2012) proposed that this success might be attributed to characteristics like discipline, motivation, and the labor market structure, which positively impact women's academic performance. Similarly, the age of entry was found to have a similar effect on dropout rates at the University of Malaga in Spain, as revealed in the study of Lassibille and Gómez (2008). This effect could be linked to factors such as students' continuity in studies after finishing high school and the freedom that comes with age to dedicate oneself exclusively to academics. Although other studies have shown a correlation between age and dropout, Aina et al. (2018) clarified that a causal relationship cannot be established since no study has controlled for the nonrandom selection of first-year students by age at enrollment.

It is particularly worth delving into one variable from the first model that remains relevant in the second model: the level of parents' education. Stratton et al. (2008) found that students whose parents have completed college are significantly more likely to remain continuously enrolled than those whose parents have lower levels of education. As mentioned earlier, Wallner and Nissen (2018) and Gury (2011) reported similar results. A higher level of parents' education can influence students' dropout risk in several ways. Firstly, higher-educated parents can offer more support and assistance to their children. Secondly, it can affect household income, as higher-educated parents generally earn higher salaries, reducing the risk of dropout due to economic problems. Lastly, students may be motivated to achieve at least the same level of education as their parents, leading to a lower dropout risk when parents have higher education levels (Aina, 2005; Bird, 2018; Farah & Upadhyay, 2017). On the other hand, the more significant impact of the mother's education may be attributed to the central role she plays in the home and society, as higher education can better guide and prepare her children for the future (Shaheen & Awan, 2020). For this reason, authors such as Marks (2008) find that the impact of the mother's education is usually more significant than or comparable to that of the father's education on students' academic performance.

As mentioned earlier, the findings offer valuable insights for institutions and policymakers to take informed actions. For instance, the charter school network Alliance uses pre-entry variables to analyze university graduation rates in the United States. They have observed that, on average, universities tend to have lower graduation rates when they enroll more low-income students, African American and

Latino students, male students, and mature students (Leonhardt & Chinoy, 2019). This information could be instrumental in shaping public policies that target specific groups statistically more prone to dropping out, ultimately facilitating higher education opportunities for these demographics.

In this regard, Tenório de Freitas and Bezerra (2022) have identified four types of policies: economic, assistance-based, academically focused, and criteria-based on students' needs/vulnerability, which can be implemented at both the governmental and higher education institution (HEI) levels. Scholarships, grants, and loans are among the most utilized policies, and they significantly impact students' decisions to persist and complete their studies (Eather et al., 2022; Tenório de Freitas & Bezerra, 2022). For example, Ison (2021) indicates that an outstanding tuition balance drastically decreases the likelihood of graduation. Governments frequently implement these types of policies.

On the other hand, the set of variables where the discussion possibilities can be broader is the institutional and academic ones. In their New York Times article on college dropout rates in the US, Leonhardt and Chinoy (2019) presents an intriguing argument, indicating that universities with similar student demographics often exhibit different dropout rates. This suggests that the issue may lie not with the students but with the institutions themselves. Some institutions actively address this problem and have succeeded in improving student performance, increasing graduation rates, and consequently reducing dropout rates, whereas others have not achieved such changes. Our study seems to support this viewpoint, as institutional and academic performance variables hold greater significance in the dropout risk.

The results show that students from newer cohorts are less likely to drop out than older cohorts. This could reflect the adoption of institutional policies to enhance retention levels and academic performance. Some of these policies include providing academic and non-academic support to students, such as tutoring systems, psychoeducational support, and teacher training. Moreover, based on these experiences, there have been discussions regarding new institutional policies related to teaching schedules, prioritizing teachers with better evaluations in the early levels, which is expected to reduce dropout rates. Despite the actions taken, it is worth considering experiences from other contexts. For example, Arias et al. (2023) suggest that measures like leveling courses, remedial courses, changes in pedagogical strategies, teacher training in effective teaching and learning methodologies, and curriculum changes or adaptations that offer schedule flexibility could be helpful too. Additionally, specific policies for first-year students, during which the highest dropout rates are recorded, are worth mentioning. Eather et al. (2022) provide examples of such actions, including foundational programs that improve students' understanding of the discipline they will study and preparatory programs or bridge courses to develop the necessary skills, knowledge, and confidence for success in university.

Another finding is that tuition discounts reduce the dropout risk, aligning with findings from Gury (2011) and Lassibille and Gómez (2008)The reduced probability of dropout due to a tuition discount is directly related to economic factors. However, it may also be associated with students' perception of the value of their education, leading them to stay in the institution, especially those who may face financial or academic challenges. Although only 7% of students did not receive any tuition aid, this area could be improved: most benefits are available when students start university, and they may be unaware of the existence of this aid or how to access it. It is important to note that PUCE's central financial aid policy (in monetary terms) is linked to household socioeconomic status, and the university's co-funded aspect aims to support students with fewer economic resources through government funding.

The exam score variable demonstrates that the skills students develop before entering university are relevant and can impact, particularly during the critical first few years, when dropout rates are highest. It should be noted that the entrance exam tests reasoning and aptitude rather than specific knowledge since it is standardized for all applicants, and each academic department determines the cutoff point for admission based on its demand. In a survey, Aina et al. (2018) confirmed that well-designed admission criteria can be a good predictor of dropout rates.

The GPA and the ratio of failed credits versus registered credits may be influenced by institutional policies that require students who fail the same course three times to leave the degree program. However, in many cases, dropout is voluntary, as students in this situation often choose to leave the university entirely. These findings align with studies conducted by Ameri et al. (2016), Da Costa et al. (2018), and Ratnaningsih et al. (2021), who also found that better grades have a positive effect on reducing dropout rates and analyzed the impact of courses completed and/or failed by students.

Authors such as Eather et al. (2022) and Neiterman et al. (2023) highlight the significance of essential policies and strategies related to peer support, including mentoring, tutoring, assisted learning, directed learning, support networks, or study groups. These strategies have demonstrated various positive outcomes in academic, instrumental, and personal aspects for students, contributing to their satisfaction and academic success, particularly for low-income and racially diverse students who gain confidence and improve mental and emotional well-being. Additionally, these approaches create a stimulating learning environment with a sense of community and shared professional identity. The study conducted by Guadalupe and Gonzalez-Gordon (2022) for Economics students at PUCE-Q shows the existence of a group effect on academic performance, and the course enrollment system at PUCE introduces bias by giving preference to high-performing students when selecting courses and schedules. This calls for a review of the current policy, considering that academic performance is a fundamental factor in dropout risk.

Furthermore, studies have demonstrated a strong relationship between students' physical and mental health and dropout rates. Therefore, implementing support services is highly recommended to address stress, anxiety, and depression because these factors impact the outcomes (Arias et al., 2023).

Conclusions

The results observed during the research confirm the complexity of analyzing dropout rates in HEIs. Thus, policies aimed at reducing dropout rates have become topics of increasing interest among researchers in recent years in the United States, Latin America, and worldwide (Tenório de Freitas & Bezerra, 2022; Arias et al., 2023; Eather et al., 2022) due to the impacts that this variable can have on educational, employment, and efficiency outcomes at the country, institutional, and individual levels.

Furthermore, the obtained results are consistent with those analyzed in previous research. However, it is essential to consider that the difference lies mainly in the characteristics of the policies to be implemented in each institution, as they must respond to the socio-cultural context in which they are developed and even to the particularities of the institution in which they are applied.

While the phenomenon of dropout has been studied using various methodologies, the application of Cox survival models, both classical and extended, used in this research allowed us to model complexities present in the study data, such as censoring, time-varying variables, and widely heterogeneous observations. The results showed that considering all these characteristics was relevant in the analysis of dropouts in the case study.

The analysis of dropout factors in a co-financed university, such as PUCE, has allowed us to infer that, given the complexity of the phenomenon studied, the current policies in place represent a starting point for addressing the issue, such as tutoring and financial assistance. However, the discussions developed in our analysis open the possibility of considering other elements as best practices when implementing institutional policies. Finally, the analysis presents examples of actions PUCE took that can serve as a reference for other contexts.

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Appendix

Given the null hypothesis is that of proportionality, if the p-value is greater than the significance level, the hypothesis is accepted and vice versa. The interest is in accepting this hypothesis so that the proportional hazards model is valid, which implies having high p-values. The calculation is performed for each variable and overall.

Table A1. Cox proportional hazards test (model 1).

	rho	chi-square	p-value
Sex	−0.075	2.91	0.09
From Pichincha province	−0.045	0.97	0.33
Age	−0.035	0.65	0.42
Unknown high school	−0.007	0.02	0.88
Private high school	0.053	1.45	0.22
Semi-public high school	0.029	0.45	0.50
Father's education level unknown	−0.057	1.92	0.17
Father's university-level education	−0.063	2.00	0.16
Mother's education level unknown	−0.010	0.06	0.81
Mother's university-level education	0.026	0.35	0.56
Total		**10.11**	0.43

Source: Prepared by the authors using university administrative records and those from the Ministry of Education and Civil Registry Office.